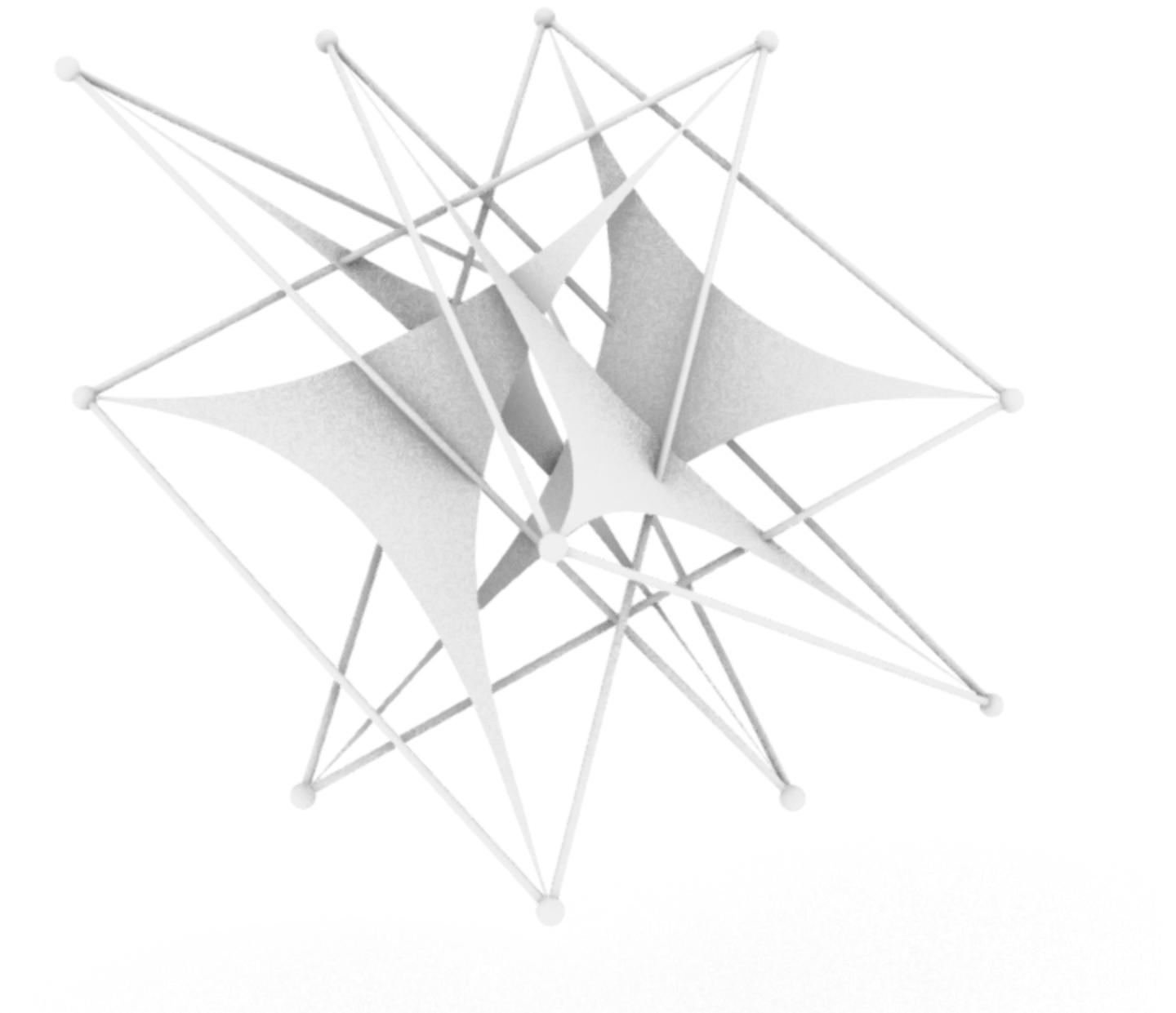


Assignment 2 - Pavilion

Jacob Pierce

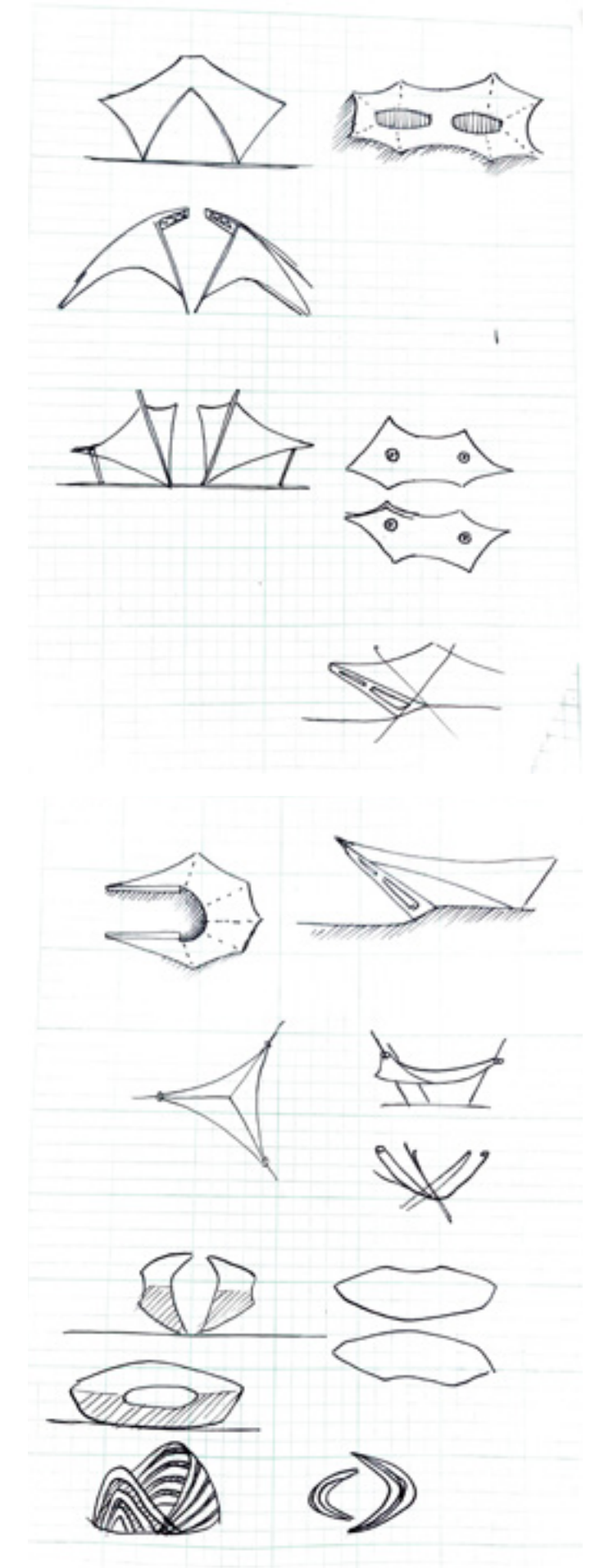
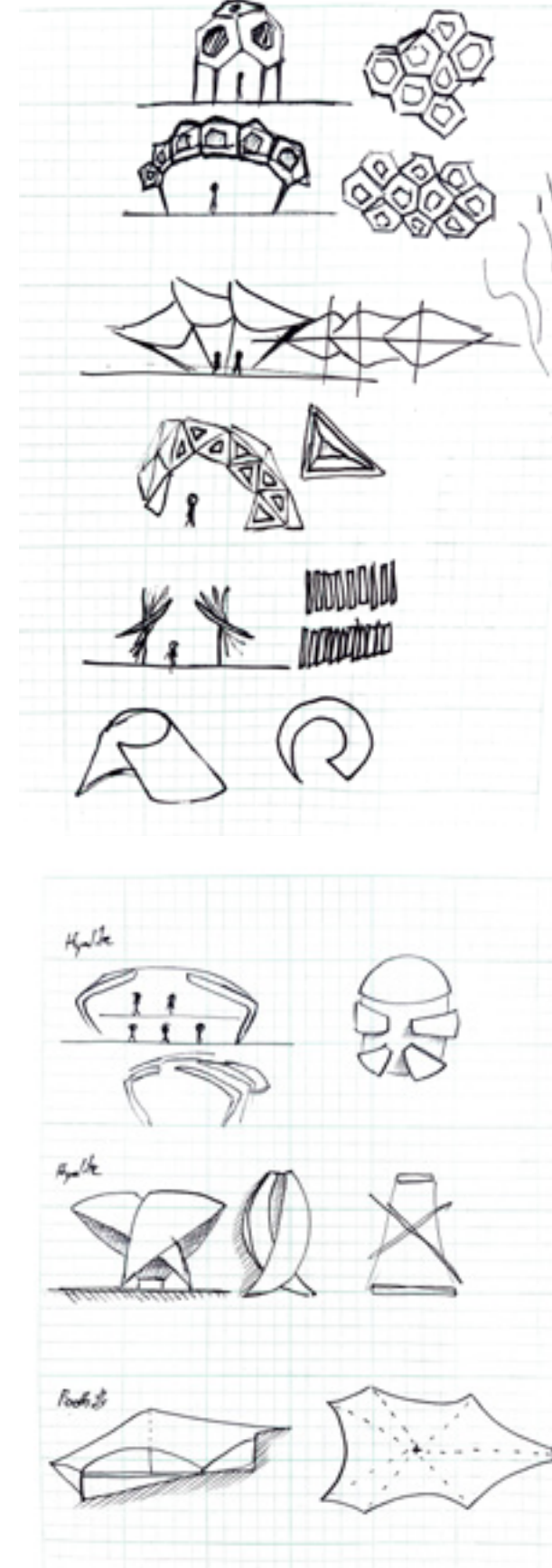
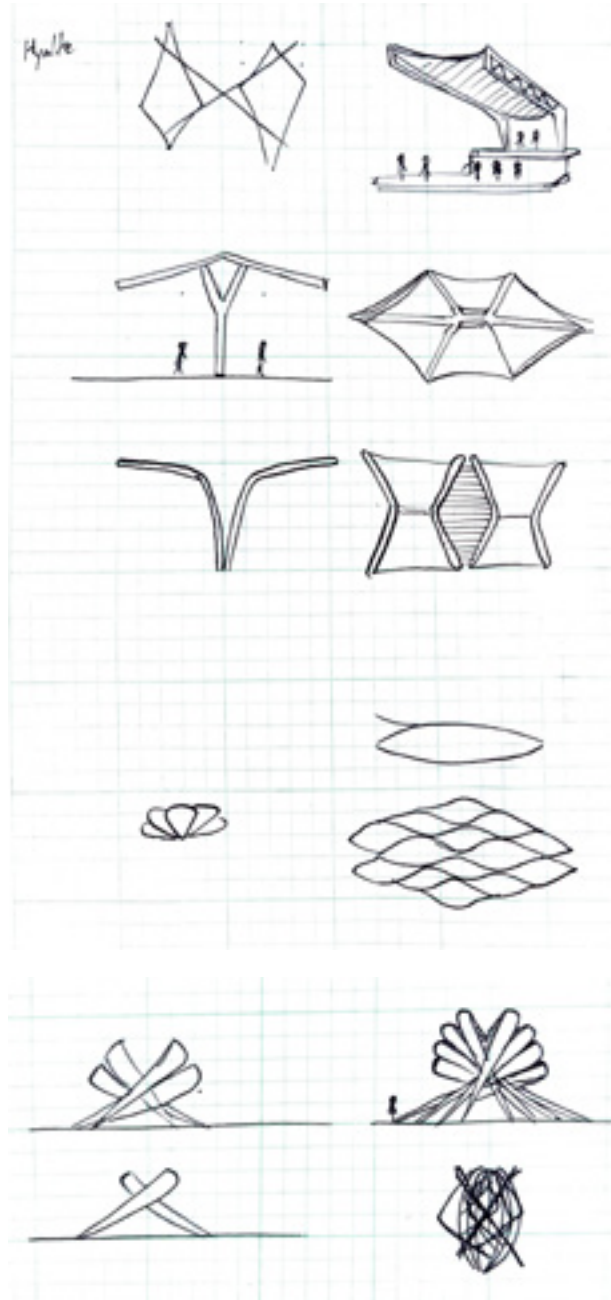
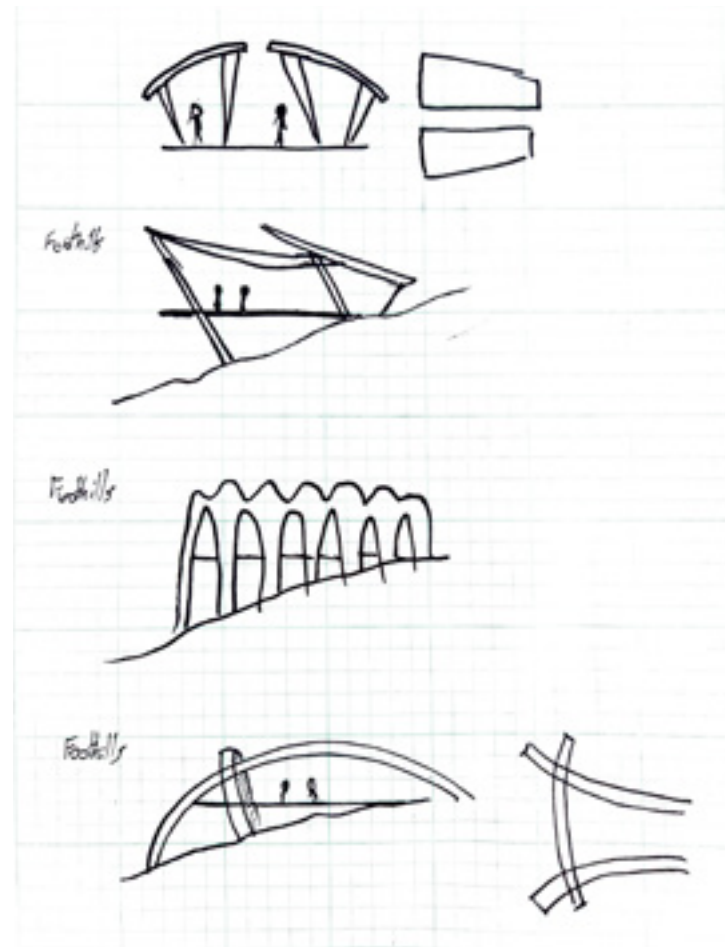
In taking this class a second time, I sought to dive into the curiosities I felt were left unexplored from last semester. While there were many of these, I wanted to explore the inherent parametricism in physics-based modeling. Using a mix of hand drawing, physical modeling (albeit poorly), and digital modeling and analysis, I was able to better understand and visualize the forces in the structures I make.



Sketching

In the beginning, I didn't really know what the direction of this class was going to be. I knew I wanted to explore options of parametric modeling, but I wasn't sure what to apply it to, as my studio class was focused on a traditional approach to construction. Due to this, I decided upon a pavilion design. Originally, it was a live music venue, but as the semester continued, I was pulled in different directions.

These sketches are the result of me exploring what the pavilion's form might look like. I was focused on catenary and tensile systems, supported by some structural frame. At the same time as I was designing the pavilion, I was also deciding on what the site might look like. Hoping to make this a local project, I alternated between Hyalite Reservoir and the Bridger foothills as my site conditions.



Physical Models

After a bit of discussion on where the next steps would lead me, the decision was made to explore physical space before a digital space. These 3 models are examples of me exploring tensile systems withing an 8" x 8" box. While my modeling skills are crude, these pieces made me think more critically about how the form responded to the limitations I was given.

Figure 1 represented my testing out the properties of the material (bending, elastic properties, connections) and lacks a cohesive formal motif.

Figure 2 resulted from me applying the information I had gained about the cloth and attempting to make a shape that was more interesting and deliberate than before.

Figure 3 was my final rendition of this tensile system. The form consists of four triangles, each connected to the frame with a vertex point and two mid points. These connections are rotated as the shape is copied around the cube. This was the shape I chose to explore in the digital realm.



Figure 1



Figure 2

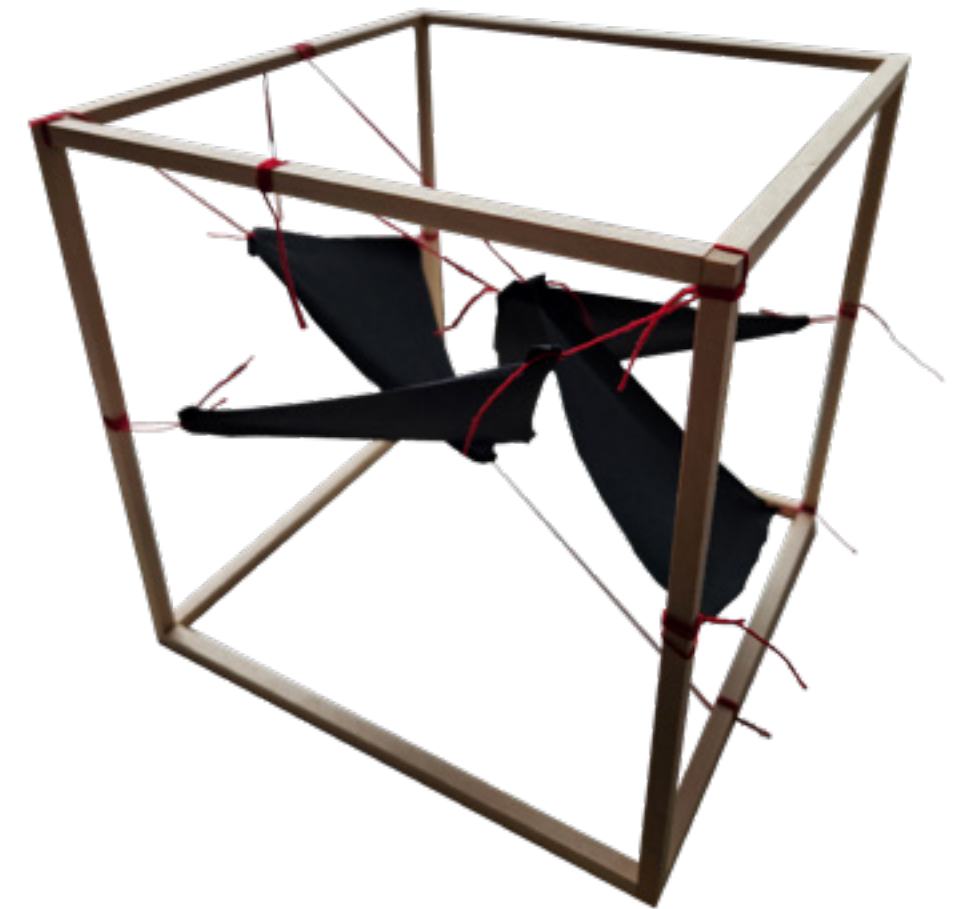
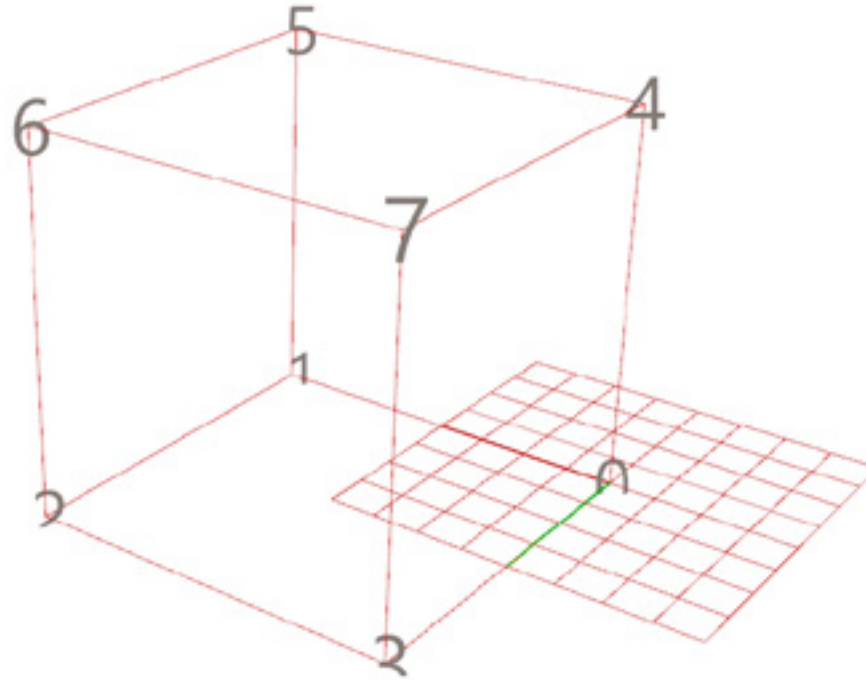


Figure 3

Digital Models

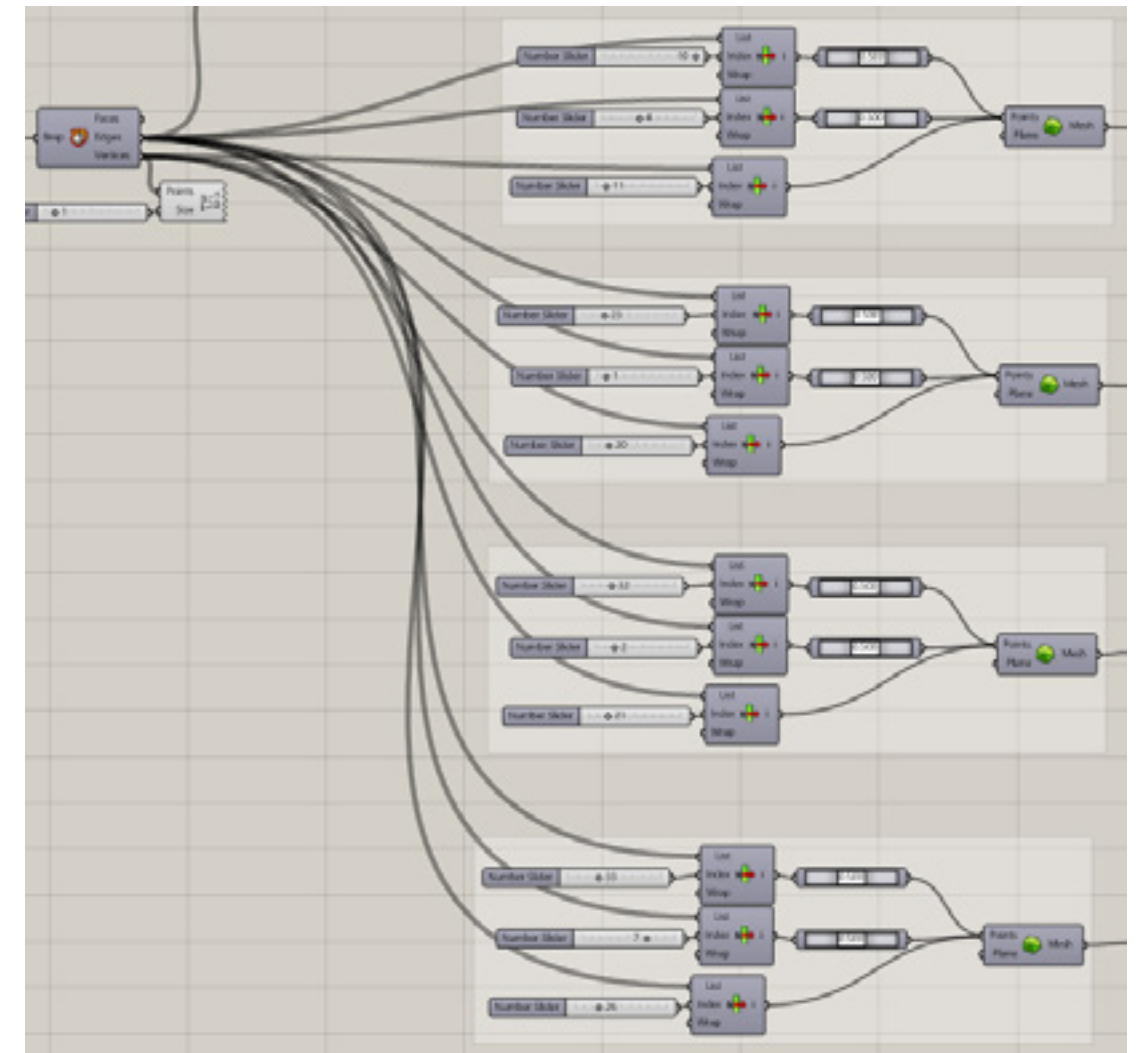
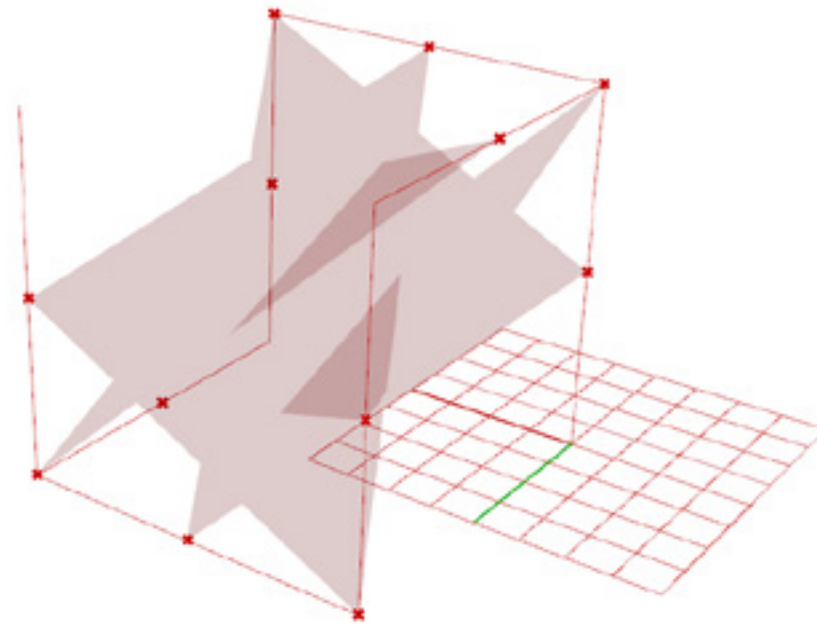
1.

The first step of recreating this model, was making the cube it was bound to. I used adjustable sliders to define a base shape and the height it should be extruded to.



2.

The second step consisted of picking out the locations at which the tensile material would connect to. As mentioned before, each triangle gets one vertex and two center points. By picking out the vertices and dividing the existing edges, I was able to extract the points I needed.



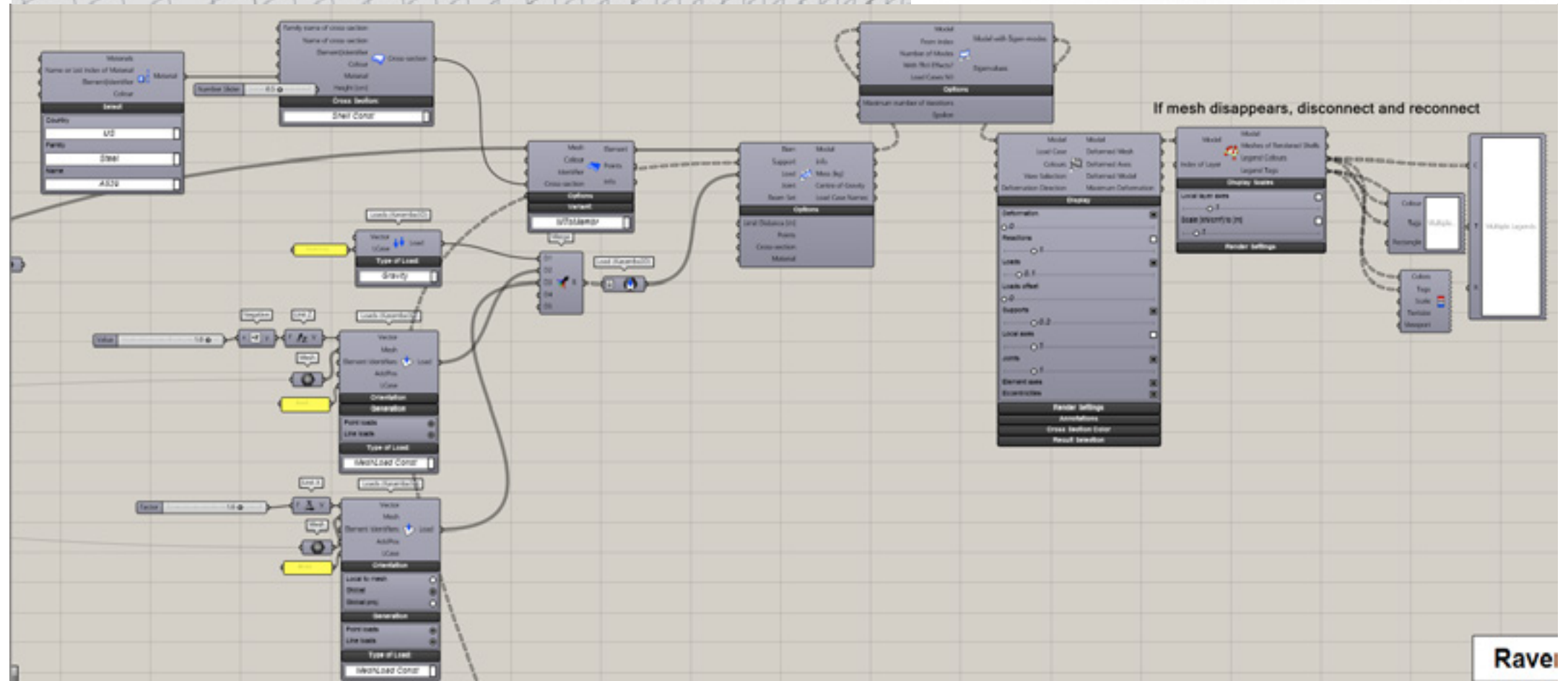
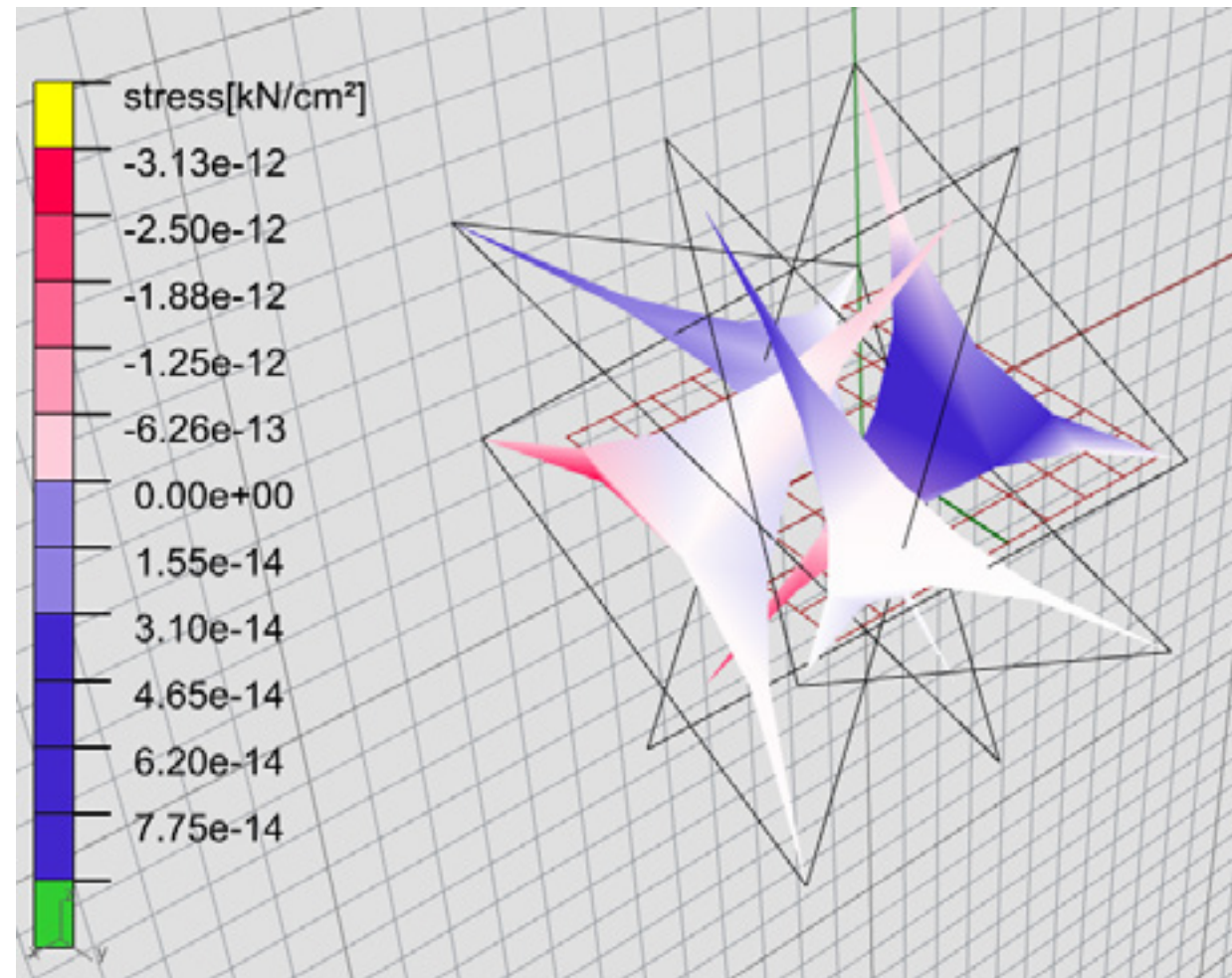
3.



Karamba

After getting a basic form, I sought out to examine it through a physics based software called Karamba3d. This created a visual representation of the forces present in each quadrant of my pavilion. It displayed digitally the physical forces I had been working with while modeling physically.

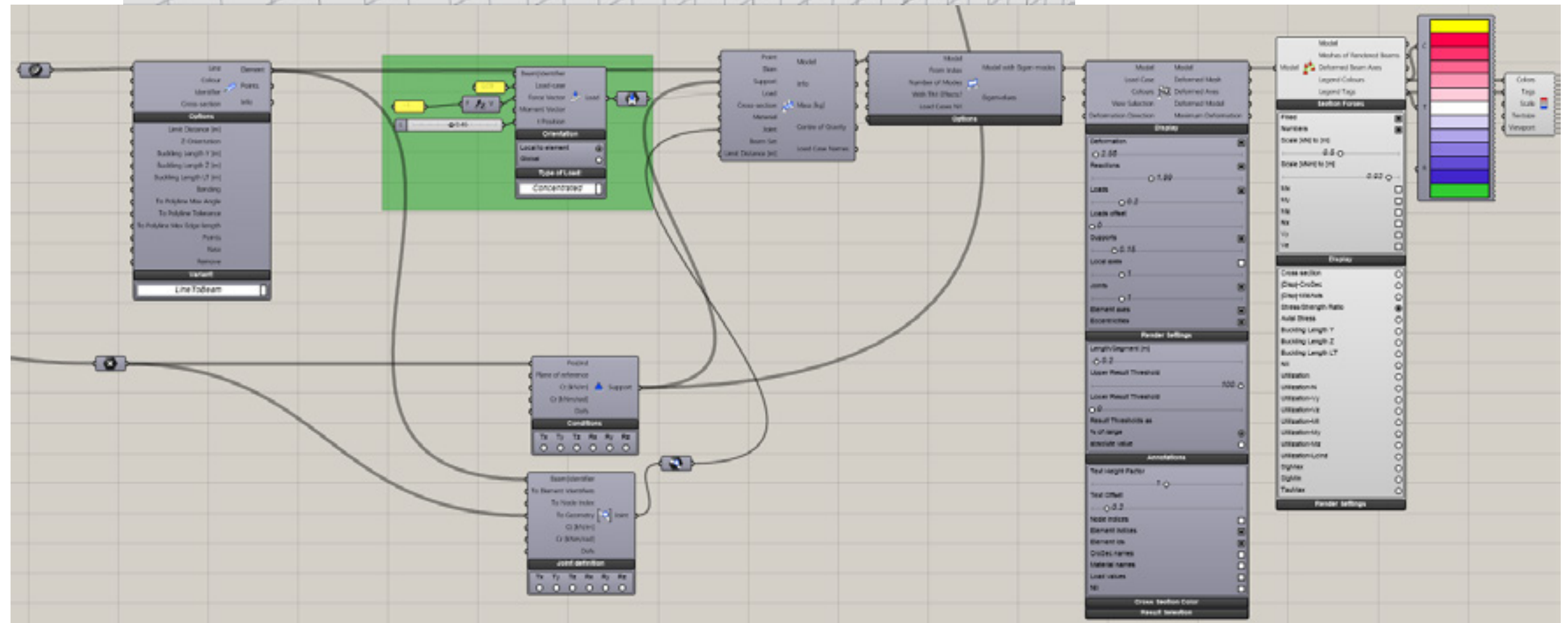
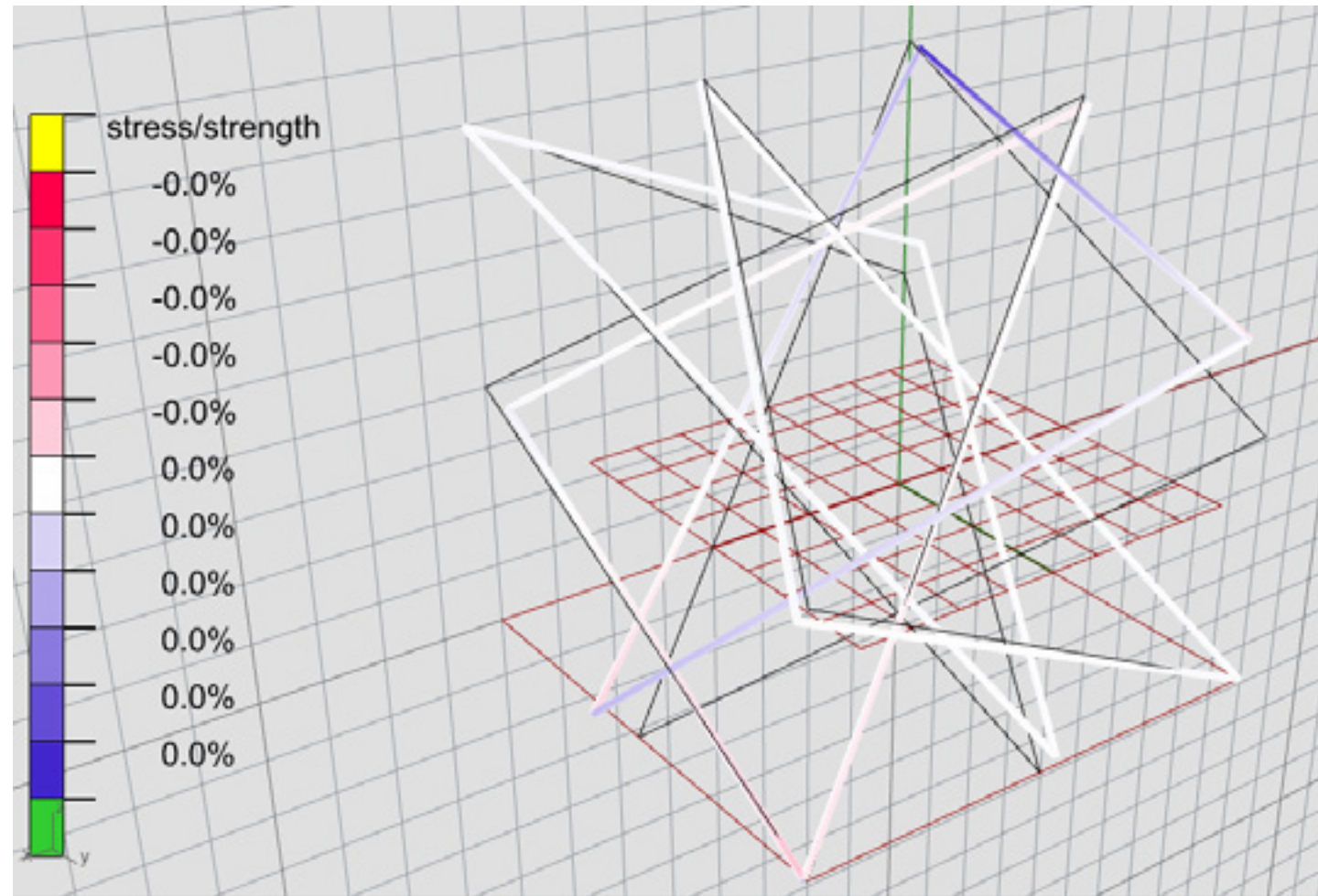
This slide shows tension forces in the tensile members in specific. Parameters such as support points, gravity load, and material properties had to be set up.



Karamba

This slide shows the stress on my structural members and the deformation they exhibit because of it. Most of the parameters remain the same, such as gravity, support points, material, etc. Basically anything needed to construct a force or load diagram.

The green portion of this script was taken from a tutorial on using the Karamba software



Unreal Rendering With Cesium



Unreal Rendering With Cesium

